

Explainable AI-Based Personalized Learning Path Recommendation System for University Students

Abhay Pratap Singh, Akshay Guleria, Uday Chaturvedi, Gursimar Makkar and Shikha Kamal

Abhay Pratap Singh, Department of Computer Science, Chandigarh University, Mohali, India. e-mail: 23BCS11784@cuchd.in
Akshay Guleria, Department of Computer Science, Chandigarh University, Mohali, India. e-mail: 23BCS10517@cuchd.in
Uday Chaturvedi, Department of Computer Science, Chandigarh University, Mohali, India. e-mail: 23BCS10807@cuchd.in
Gursimar Makkar, Department of Computer Science, Chandigarh University, Mohali, India. e-mail: 23BCS11455@cuchd.in
Shikha Kamal, Department of Computer Science, Chandigarh University, Mohali, India. e-mail: shikha.e12552@cumail.in

Abstract Digital learning platforms generate large volumes of student interaction data, yet educational recommender systems offer little transparency, rarely model forgetting, and are seldom evaluated in a way that shows which components actually contribute. This chapter proposes an Explainable AI (XAI) based learning path recommender combining a knowledge-graph-enhanced Deep Knowledge Tracing module, a prerequisite-constrained planner and a priority-backtracking explanation module. On the Open University Learning Analytics Dataset (OULAD), under five-fold cross-validation grouped by student, it attains AUC-ROC 0.9538 ± 0.0019 and RMSE 0.2621 ± 0.0026 , exceeding DKT ($p = 0.0009$) and a GNN baseline ($p = 0.022$) after Holm correction. The central finding concerns evaluation: removing the knowledge graph changes accuracy by only -0.0001 ($p = 0.77$), because accuracy is scored at assessment positions and only 89 of 237 concepts carry assessments. Under a cold-concept protocol that withholds assessments for a subset of concepts and scores only those, the same graph improves accuracy by $+0.0325$ ($p = 0.0003$, Cohen's $d = 5.06$), winning on every fold. The planner makes zero prerequisite violations across 906 decisions evaluated against real learner behaviour, and explanations are assessed by objective fidelity, sufficiency and stability rather than subjective rating.

Keywords Explainable AI · Personalized learning path · Knowledge tracing · Knowledge graphs · Educational data mining · Ablation methodology

1 Introduction

Higher education's embrace of digital learning has opened new possibilities for individualized instruction, yet conventional courses still follow one path for all students despite differences in prior knowledge and pace, causing course mis-selection and knowledge gaps [1–3]. Personalized learning path recommendation can address this, but most systems concentrate on prediction accuracy and offer little explanation for any recommendation [4]. Designing a system that describes a learner's evolving knowledge, sequences activities well and explains its choices is therefore a major challenge. Knowledge tracing methods assess learning rather than optimize progression [5, 6], automated sequencing remains opaque to students and advisers [7, 8], and the link between knowledge graphs, sequential decision-making and temporal modeling is underexplored in education [10, 11].

A further limitation motivates this work. Such systems combine several components, each justified by an ablation showing that its removal degrades accuracy. When a structural prior exists to support concepts whose supervision is scarce, but accuracy is measured only where supervision is abundant, the ablation tests the component in the one regime where it cannot matter. The contributions of this work are:

- A knowledge-graph-enhanced knowledge tracing model with a learned per-head temporal decay, attaining AUC-ROC 0.9538 and calibration error 0.0054 on OULAD.
- Evidence that conventional ablation cannot detect a knowledge graph's contribution, and a cold-concept protocol that can ($+0.0325$, $d = 5.06$).
- A prerequisite-constrained planner whose pedagogical guarantee is enforced by construction, verified at zero violations across 906 decisions.
- Objective explanation metrics (fidelity, sufficiency, stability) in place of subjective rating, released with a reproducible pipeline.

2 Literature Review

Early educational recommenders used content-based and collaborative filtering, which suffer cold-start problems and data sparsity [1, 2]; hybrid models improved robustness but forecast preferences rather than modeling knowledge acquisition [4]. Deep Knowledge Tracing (DKT) [5] learns the latent knowledge state directly from interaction data, unlike Bayesian Knowledge Tracing's static parameters [6], and attention-based successors [21, 7] attend to any earlier interaction [9]. Reinforcement learning frames path recommendation as a Markov Decision Process and, combined with knowledge tracing, has been reported to reduce dropout [8, 18]. Knowledge graphs support structured, explainable recommendation by modelling concepts and their dependencies [10, 15], with Bloom's taxonomy providing a theoretical foundation for path design through six progressive cognitive levels [12], and graph neural networks handling the relational structure such graphs induce. Multi-objective hierarchical frameworks have optimized dropout risk, learner satisfaction and completion rates, adding priority-backtracking interpreters for interpretability [14, 13], and hybrid neuro-symbolic and generative approaches have begun to explore understandable explanation generation [16].

2.1 Research Gap

Four gaps remain. Explainability is added as an afterthought [17], and SHAP-based explanations dominate in education but are applied mainly at the global level rather than to individual recommendations [18]. Where explanations are evaluated at all, it is usually by participant rating, which measures whether readers found an explanation agreeable rather than whether it is faithful to the model. Many frameworks ignore the dynamic evolution of knowledge, including forgetting, which compromises the pedagogical integrity of suggested trajectories [19]. Knowledge graphs are rarely applied to sequential decision-making and explanation [20]. Finally, the contribution of such structural priors is established by ablation on aggregate accuracy, without checking whether that metric can observe the component at all.

Together these gaps limit customization, transparency and effective learning-path optimization for university students. Existing systems rarely integrate knowledge graphs, temporal knowledge tracing, constrained path optimization and pedagogically meaningful explanation in one framework, and the validation of such systems reports component contributions on metrics that may be unable to detect them, and explanation quality by opinion rather than measurement.

3 Methodology

3.1 Overall Framework

The system estimates a learner's changing knowledge state from interaction history; the prerequisite relationships of a knowledge graph then constrain a planner that selects the next learning task; finally each recommendation receives a pedagogical explanation from a priority-backtracking module [15] (Fig. 1). The framework therefore has three components: learner knowledge estimation, sequential path optimization and explanation production [5]. Each is evaluated separately, because a gain in one does not imply a gain in another: better prediction need not yield better paths, and a faithful explanation need not be a complete one.

[fig1.png: place the figure here]

Fig. 1 Overall XAI-based personalized learning recommendation framework

Table 1 Dataset and derived concept graph

Quantity	Value
Enrolments / distinct learners	32,593 / 28,785
Clickstream events	10,655,280
Assessment records	173,912
Label balance (success / not)	67.9% / 32.1%
Concepts / prerequisite / co-requisite edges	237 / 724 / 302
Concepts carrying assessments	89 of 237
Week-placement agreement (Spearman)	$\rho = 0.975$

3.2 Data Acquisition and Concept Graph

Evaluation uses OULAD [3]: 32,593 enrolments, 10,655,280 clickstream events and 173,912 assessment records across seven modules. Records are ordered

chronologically, categorical attributes one-hot encoded and numerical attributes standardized, and missing assessment scores are median-imputed only where a submission exists. Timestamps are day-granular with no intra-day resolution, so a session is one learner-day rather than an inactivity threshold. The processed data is split 80:10:10.

OULAD supplies no concept annotations or prerequisites, so the graph is derived. A concept is a (module, week-of-study) pair. Week placement uses the week_from field where present and otherwise the median access day; where both exist they agree at Spearman $\rho = 0.975$. Edges link consecutive weeks, link the four preceding weeks to each assessment week, and are mined from behaviour, accepted only where ordering is consistent and later success is measurably higher:

$$\text{prec}(a, b) = |\{u : \tau_u(a) < \tau_u(b)\}| / |\{u : a, b \in h_u\}|, \quad (1)$$

$$\text{lift}(a, b) = P(y_b | a < b) / P(y_b), \quad (2)$$

where $\tau_u(\cdot)$ is learner u 's first interaction with a concept, with support above 100 learners, $\text{prec} \geq 0.75$ and $\text{lift} > 1.1$. Requiring lift removes pairs separated only by the academic calendar. The result is 237 concepts, 724 prerequisite and 302 co-requisite edges, acyclic without intervention.

Two properties of the data matter for interpretation. OULAD records only submitted assessments: binarising at the pass mark gives a 95.6% positive rate, with the fifth percentile of submitted scores equal to the pass mark itself, and 150,013 of 323,925 expected submissions never occurred. Treating non-submission as failure gives a 67.9/32.1 balance and makes the target successful completion. Second, assessments exist for only 89 of the 237 concepts; the remaining 148 are reached only through the graph.

3.3 Knowledge Graph-Enhanced Knowledge Tracing

A Transformer [9] encodes the learner's interaction sequence, while a Graph Convolutional Network [10] encodes prerequisite relationships into concept embeddings recomputed at every forward pass, so gradients reach the graph structure (Fig. 2). The combined representation lets the model account for both the learner's past progress and the structural links between concepts. Forgetting is modelled as an attention bias that decays with elapsed time,

[fig2.png: place the figure here]

Fig. 2 Knowledge graph enhanced knowledge tracing

$$b^{(h)}_{ij} = -\text{softplus}(\gamma_h) \log(1 + \Delta_{ij}), \quad (3)$$

where Δ_{ij} is the days between interactions i and j and γ_h is learned per head; $\gamma_h = 0$ gives the temporal ablation. Clickstream events update the hidden state without contributing loss, and binary cross-entropy is computed only at assessment positions, which makes a length-200 sequence model trainable when learners

average 8.7 assessments. Because mastery is shown to learners, probabilities are calibrated by temperature scaling [22], fitted on half of each held-out fold.

3.4 Prerequisite-Constrained Path Optimization

Recommendation is formulated as a Markov Decision Process whose state holds the mastery vector, concepts already recommended, learner features and remaining budget, and the reward combines learning progress, prerequisite satisfaction, a redundancy penalty and a step cost. Rather than expecting a policy to learn the pedagogical constraint, it is encoded in the admissible actions: a concept qualifies only if the learner has not mastered it, every prerequisite is at the readiness threshold, and it belongs to an enrolled module, so an unprepared recommendation is impossible by construction (Fig. 3).

[fig3.png: place the figure here]

Fig. 3 Prerequisite-constrained learning path optimization

The evaluated policy is greedy. Each admissible concept is simulated as studied and compared against an idle period of equal length, and the concept with the largest total improvement in log-odds is chosen. The idle baseline is essential because studying advances the clock and Eq. (3) then decays every earlier interaction; log-odds are needed because engaged learners saturate near 0.98. Thresholds are the 70th and 30th percentiles of each learner's own mastery, since mastery is bimodal. A learned policy is left to future work.

3.5 Explainable Recommendation

The priority-backtracking interpreter traverses the chosen concept's prerequisites, scoring each by its shortfall below readiness, discounted by graph distance (Fig. 4). Counterfactuals raise a gap concept's mastery and re-evaluate the planner's score, so the claimed effect of closing a gap is measured rather than asserted. An unmet prerequisite immediately preceding the recommendation therefore outranks an equal shortfall further back, and every clause of the generated explanation traces to a number the model produced.

[fig4.png: place the figure here]

Fig. 4 Explainable AI module for recommendation justification

4 Implementation

4.1 Development Tools and Computing Environment

The framework is implemented in Python 3.11 with PyTorch. Data preparation runs in SQL on an embedded DuckDB database so every transformation is auditable; NetworkX builds the graph and FP-Growth (mlxtend) mines co-requisites. Training used one NVIDIA T4 GPU and all inference runs on CPU; data preparation takes about 2.5 minutes and training one configuration about 25 minutes.

4.2 Dataset Configuration

A pipeline built on SQL, Pandas and NumPy turns the raw records into per-enrolment sequences, 27,904 of them from 25,101 learners, with 242,670 supervised assessment positions (Table 1). Categorical attributes such as gender and education level are one-hot encoded, and long-tailed click counts are log-transformed before standardization so a few very active students do not dominate

the scale. Sequences are windowed to the most recent 200 interactions. Cross-validation folds are stratified on final outcome, and no learner appears in more than one fold.

4.3 Model Implementation Specifications

The encoder has two Transformer layers, hidden size 128, four heads and sequence length 200; the three GCN layers use layer normalization and ReLU, giving 64-dimensional concept embeddings. Training uses Adam (learning rate 10^{-3} , weight decay 10^{-4} , cosine annealing), dropout 0.2, batch size 64 and early stopping at patience 3. The model has 466,569 parameters, and one batched pass over all admissible concepts keeps an explained recommendation near 160 ms on CPU. Random seeds are fixed and recorded with every result.

4.4 Evaluation and Validation

Baselines are a majority-class predictor, SAKT, DKT and a GNN-based recommender. Predictive quality uses AUC-ROC, RMSE and expected calibration error. Path quality is measured against observed behaviour: held-out histories are truncated, each planner recommends, and ground truth is the concepts the learner actually studied next, with only learners who passed or achieved distinction contributing ground truth. Explanation quality is measured by fidelity, sufficiency and stability. Ablations remove the knowledge graph and the temporal decay, under both the conventional protocol and the cold-concept protocol of Sect. 5. Cross-validation is five-fold and grouped by student; comparisons use paired t-tests with Holm–Bonferroni correction [23], with Cohen's d reported beside every p-value, since with many paired decisions significance is attainable at negligible effect size.

5 Results and Discussion

Table 2 Predictive performance and component ablation. Folds grouped by student, p Holm-corrected. Proposed expected calibration error = 0.0054

Model	AUC-ROC	RMSE	p
Majority (no-skill)	0.5000 $\pm$ 0.0000	0.4669	<0.001
SAKT	0.9233 $\pm$ 0.0018	0.2992	<0.001
DKT	0.9435 $\pm$ 0.0017	0.2751	0.0009
GNN-based	0.9502 $\pm$ 0.0014	0.2665	0.0220
Proposed	0.9538 $\pm$ 0.0019	0.2621	—
w/o knowledge graph	0.9539 $\pm$ 0.0021	0.2618	0.7704
w/o temporal attention	0.9532 $\pm$ 0.0014	0.2624	0.4507

5.1 Predictive Accuracy

As Table 2 shows, the proposed model attains the highest AUC-ROC (0.9538 $\pm$ 0.0019) and lowest RMSE (0.2621), exceeding DKT by 0.0103 ($p = 0.0009$) and the GNN baseline by 0.0036 ($p = 0.022$). The majority baseline scores exactly 0.5000, so the result is not an artefact of class imbalance, and SAKT reaches only

0.9233, so the gain is not simply from attention. Calibration error falls from 0.0084 to 0.0054 after temperature scaling (Fig. 5). Because predicted mastery is shown to learners inside explanations, a poorly calibrated probability would be an incorrect statement rather than a cosmetic defect.

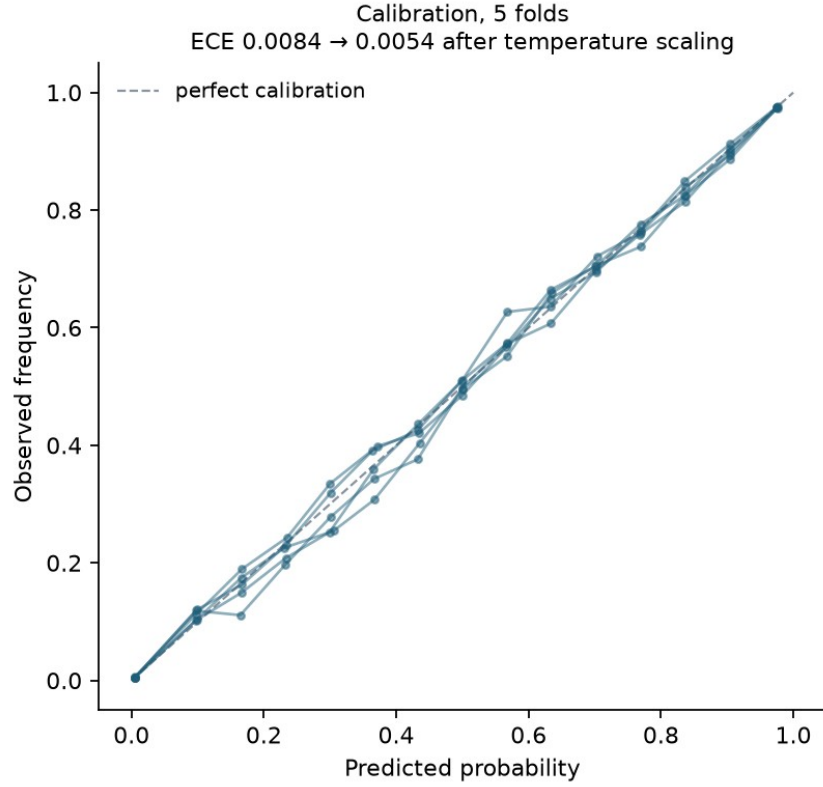


Fig. 5 Calibration of predicted mastery across the five folds. Expected calibration error falls from 0.0084 to 0.0054 after temperature scaling

Table 3 Learning path quality over 906 decisions from 400 held-out learners. Effect size against the proposed planner: syllabus order $d = 0.36$, random $d = 0.09$; all $p < 0.02$

Planner	NDCG@5	Hit@5	Viol.	Cov.
Syllabus order	0.0612	0.1832	0.0000	0.452
Popularity	0.1069	0.2936	0.0000	0.510
Random (admissible)	0.1300	0.3455	0.0000	0.722
Weakest-first	0.1344	0.3642	0.0000	0.515
Proposed	0.1532	0.3687	0.0000	0.726

5.2 Learning Path Quality

Across 906 decisions from 400 held-out successful learners (Table 3), the proposed planner achieves the highest NDCG@5 (0.1532) and Hit@5 (0.3687). All differences are significant, but effect sizes qualify this: $d = 0.36$ against syllabus order and only $d = 0.09$ against random admissible selection. The prerequisite constraint does most of the work, with the learned ranking adding a small consistent gain. Violations are zero for every planner, and coverage of 0.726 shows the planner does not collapse onto popular concepts, the failure mode popularity-based ranking shows at 0.510. The small margin over random selection also bounds the headroom for a learned policy.

5.3 Explanation Quality

Explanations were evaluated objectively rather than by participant rating. The sample is small, 25 decisions with three perturbations each, but every measure is computed against the model's own counterfactual response rather than an opinion. Fidelity is 0.5318 and stability 0.5310: cited gaps are positively but imperfectly predictive of measured improvement, and about half persist under perturbation. Sufficiency of 0.0020 is a negative finding. The named concept accounted for only 0.2% of predicted benefit, the rest arising from unmentioned related concepts, so explanations were accurate but incomplete, and the renderer was amended to name the main beneficiaries. No rating scale would reveal this, because participants cannot judge a quantity the explanation omits.

Table 4 Cold-concept ablation. Assessments withheld for 27 of the 89 assessed concepts and scored on those only

Variant	AUC-ROC	RMSE
With knowledge graph	0.9004 $\pm$ 0.0059	0.3359
Without	0.8679 $\pm$ 0.0042	0.3611
Difference	+0.0325, $p = 0.0003$, $d = 5.06$	

5.4 Why Conventional Ablation Is Insufficient

Removing the knowledge graph changes accuracy by -0.0001 ($p = 0.77$) and removing temporal attention by -0.0006 ($p = 0.45$) (Table 2). Read at face value, neither contributes. But accuracy is computed only at assessment positions, covering 89 concepts with about 2,700 labelled examples each, where a free embedding learns each concept adequately without any structural information, so the graph is genuinely redundant there. The graph exists for the other 148 concepts, which never enter the metric.

To measure it directly, all assessments were withheld for 27 of the 89 assessed concepts and their responses removed from the input; evaluation was restricted to those concepts. Table 4 shows 0.9004 with the graph against 0.8679 without, a gain of $+0.0325$ ($p = 0.0003$, $d = 5.06$) on every fold. The graph contributes nothing where supervision is plentiful and substantially where it is absent, which matters because the recommender ranks across all 237 concepts. Component ablations should be accompanied by evidence that the metric can detect the component removed.

5.5 Overall Discussion

Combining knowledge graph modeling, temporal knowledge tracing, constrained planning and explanation improves both predictive and recommendation metrics over the baselines. Explanation adds computation, about 160 ms for a fully explained recommendation against 7 ms for a cached mastery lookup, which remains well within the latency of an interactive advising interface.

6 Conclusion and Future Work

6.1 Conclusion

This study combined knowledge graph modeling, Deep Knowledge Tracing, prerequisite-constrained planning and priority-backtracking explanation into an explainable learning path recommender. On OULAD it exceeded majority, SAKT, DKT and GNN baselines (AUC-ROC 0.9538, calibration error 0.0054) and made no prerequisite violations. Its principal finding is methodological: a knowledge graph that conventional ablation reports as useless improves cold-concept accuracy by +0.0325 with $d = 5.06$. The discrepancy is a property of the metric, not the model: aggregate accuracy measured where supervision is abundant cannot observe a structural prior whose purpose is to compensate for its absence. Verifying such systems therefore requires evaluation protocols matched to what each component is for.

6.2 Limitations

The prediction target is successful completion rather than knowledge. Because OULAD records only submitted assessments, treating non-submission as failure is necessary for a usable class balance, but a substantial share of the signal is then engagement rather than understanding: appropriate for a system intended to prompt intervention, but a weaker claim than measuring mastery. The concept graph is likewise derived rather than given. OULAD supplies no concept annotations or material titles, placement for 82% of materials relies on observed access, and mining prerequisites from the same behaviour that orders the concepts is partially self-referential.

Supervision covers 89 of 237 concepts, so mastery for the remainder is not directly calibrated, and predicted mastery is bimodal, which is why learner-relative thresholds are needed. The proposed model uses learner features that the DKT and SAKT baselines, following their original formulations, do not, so part of the gain may reflect feature access rather than architecture. Finally, all data comes from a single institution's distance-learning courses, and generalization to campus-based teaching is untested.

6.3 Ethical Considerations

OULAD is publicly released under CC-BY 4.0 and anonymised, and no data was collected from participants. The system is advisory. Because the model conditions on gender, disability, deprivation band and age band, it should not inform assessment or progression decisions until a subgroup fairness analysis has been carried out.

6.4 Future Work

Four directions follow from the limitations. The most immediate is a fairness analysis: subgroup performance across gender, disability, deprivation band and age band must be established before any deployment. Second, a learned policy over the same state and reward should be trained and evaluated, in simulation and by off-policy estimation on logged trajectories; algorithms such as Soft Actor-Critic or Proximal Policy Optimization would need to beat a greedy planner already $d = 0.09$ above random selection to justify their complexity. Third, a user evaluation would address what the objective measures cannot: fidelity, sufficiency and stability show whether explanations are faithful to the model, not whether learners find them useful. Fourth, expert validation of the mined edges would give an independent estimate of graph precision. Beyond these, integrating forum and video-lecture activity, testing transfer across subjects and institutions, and building lighter models through pruning and distillation are natural extensions.

Acknowledgements All source code, SQL transformations and results are publicly available, and every table in this chapter is regenerated from the raw data by a single command with fixed random seeds:

<https://github.com/abhay27singh/explainable-learning-path-recommender>

References

1. Li, A., Li, Y., Gao, X.: Personalized learning path recommendation based on knowledge graphs: a survey. *Electronics* 15(1), 238 (2026)
2. Aher, S.B., Lobo, L.M.R.J.: Combination of machine learning algorithms for recommendation of courses in e-learning system based on historical data. *Knowl.-Based Syst.* 51, 1–14 (2013)
3. Kuzilek, J., Hlosta, M., Zdrahal, Z.: Open University Learning Analytics dataset. *Sci. Data* 4, 170171 (2017)
4. Corbett, A.T., Anderson, J.R.: Knowledge tracing: modeling the acquisition of procedural knowledge. *User Model. User-Adapt. Interact.* 4(4), 253–278 (1994)
5. Piech, C., Bassen, J., Huang, J., Ganguli, S., Sahami, M., Guibas, L., Sohl-Dickstein, J.: Deep knowledge tracing. In: *Advances in Neural Information Processing Systems*, vol. 28, pp. 505–513 (2015)
6. Zhang, J., Shi, X., King, I., Yeung, D.-Y.: Dynamic key-value memory networks for knowledge tracing. In: *Proceedings of the 26th International Conference on World Wide Web*, pp. 765–774 (2017)
7. Ghosh, A., Heffernan, N.T., Lan, A.S.: Context-aware attentive knowledge tracing. In: *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 2330–2339 (2020)
8. Shen, S., Liu, Q., Chen, E., Huang, Z., Huang, W., Yin, Y., Su, Y., Wang, S.: Learning process-consistent knowledge tracing. In: *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, pp. 1452–1460 (2021)

9. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, L., Polosukhin, I.: Attention is all you need. In: *Advances in Neural Information Processing Systems*, vol. 30, pp. 5998–6008 (2017)
10. Kipf, T.N., Welling, M.: Semi-supervised classification with graph convolutional networks. In: *International Conference on Learning Representations* (2017)
11. Mnih, V., Kavukcuoglu, K., Silver, D., Rusu, A.A., Veness, J., Bellemare, M.G., Graves, A., Riedmiller, M., Fidjeland, A.K., Ostrovski, G., et al.: Human-level control through deep reinforcement learning. *Nature* 518, 529–533 (2015)
12. Khosravi, H., Buckingham Shum, S., Chen, G., Conati, C., Tsai, Y.-S., Kay, J., Knight, S., Martinez-Maldonado, R., Sadiq, S., Gašević, D.: Explainable artificial intelligence in education. *Comput. Educ.: Artif. Intell.* 3, 100074 (2022)
13. Ribeiro, M.T., Singh, S., Guestrin, C.: “Why should I trust you?” Explaining the predictions of any classifier. In: *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 1135–1144 (2016)
14. Lundberg, S.M., Lee, S.-I.: A unified approach to interpreting model predictions. In: *Advances in Neural Information Processing Systems*, vol. 30, pp. 4765–4774 (2017)
15. Zhang, S., Liu, Q., Chen, Y., Wang, L., Zhao, H.: A fine-grained and multi-context-aware learning path recommendation model over knowledge graphs for online learning communities. *Inf. Process. Manag.* 60(5), 103464 (2023)
16. Yang, Y., Peng, X., Chen, M., Liu, S.: An explainable graph-based course recommendation model based on multiple interest factors. *Expert Syst. Appl.* 264, 125889 (2025)
17. Xie, X., Feng, X.: Personalized learning path recommendation based on learner profile and knowledge graph. *Comput. Inform.* 44(4), 983–1008 (2025)
18. Zhao, J., Wang, H., Li, M., Sun, Y.: Personalized learning path recommendation for middle school students based on joint optimization of deep reinforcement learning and knowledge tracing. *Sci. Rep.* (2026)
19. Boujmiraz, S., Darhmaoui, H., Drissi El Maliani, A.: Predicting student performance: a comprehensive review of machine learning, deep learning, and explainable AI approaches. *Comput. Educ.: Artif. Intell.* 10, 100548 (2026)
20. Qiang, M., Liu, Z., Zhang, R.: Explainable AI in education: integrating educational domain knowledge into the deep learning model for improved student performance prediction. *Sci. Rep.* 16, 9515 (2026)
21. Pandey, S., Karypis, G.: A self-attentive model for knowledge tracing. In: *Proceedings of the 12th International Conference on Educational Data Mining*, pp. 384–389 (2019)
22. Guo, C., Pleiss, G., Sun, Y., Weinberger, K.Q.: On calibration of modern neural networks. In: *Proceedings of the 34th International Conference on Machine Learning*, pp. 1321–1330 (2017)
23. Holm, S.: A simple sequentially rejective multiple test procedure. *Scand. J. Stat.* 6(2), 65–70 (1979)